Weak (Proxy) Factors Robust Hansen-Jagannathan Distance For Linear Asset Pricing Models

Lingwei Kong*

*Tinbergen Institute & University of Amsterdam

Co-movements between returns and factors

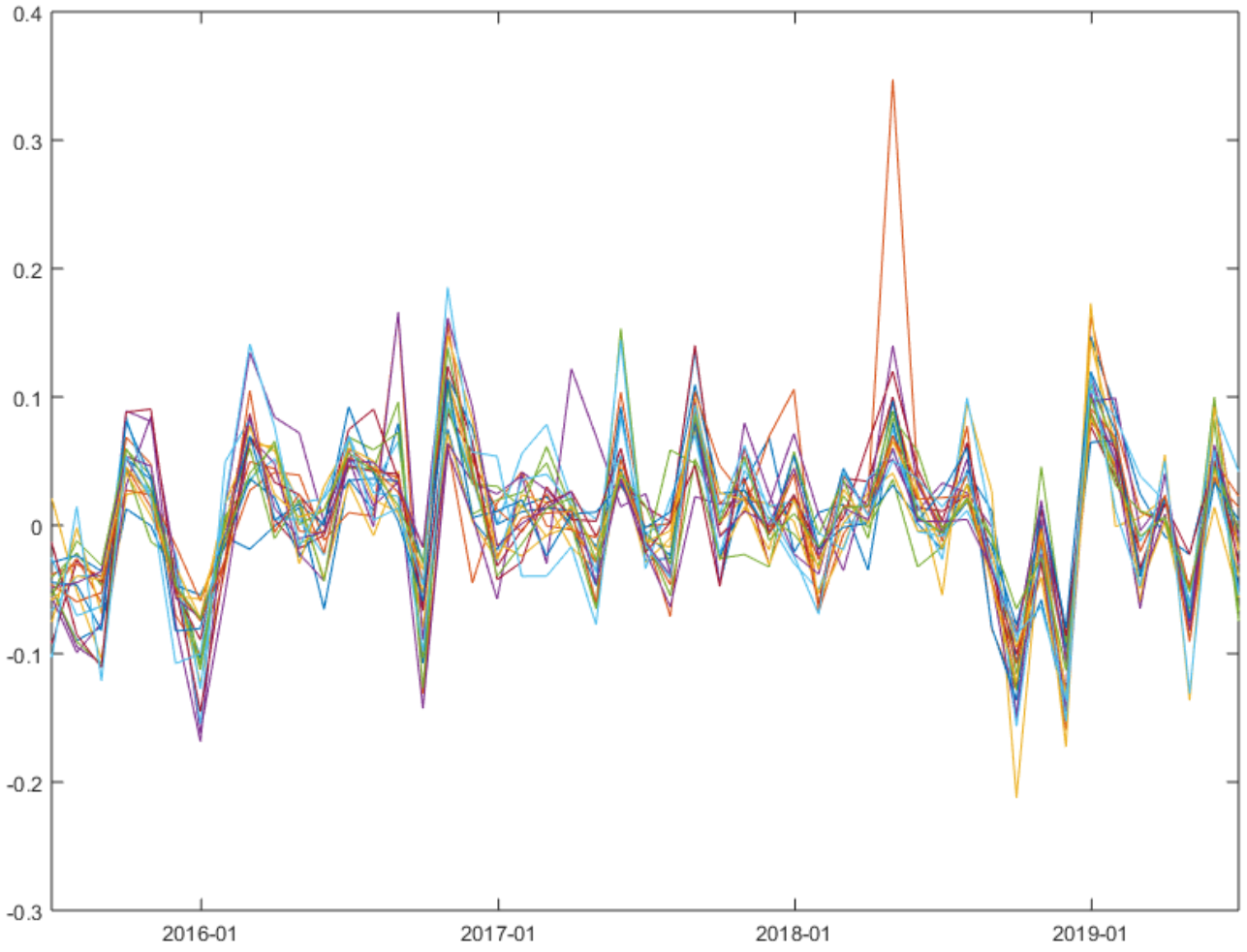

Figure: Fama-French Portfolio Returns

Co-movements between returns and factors

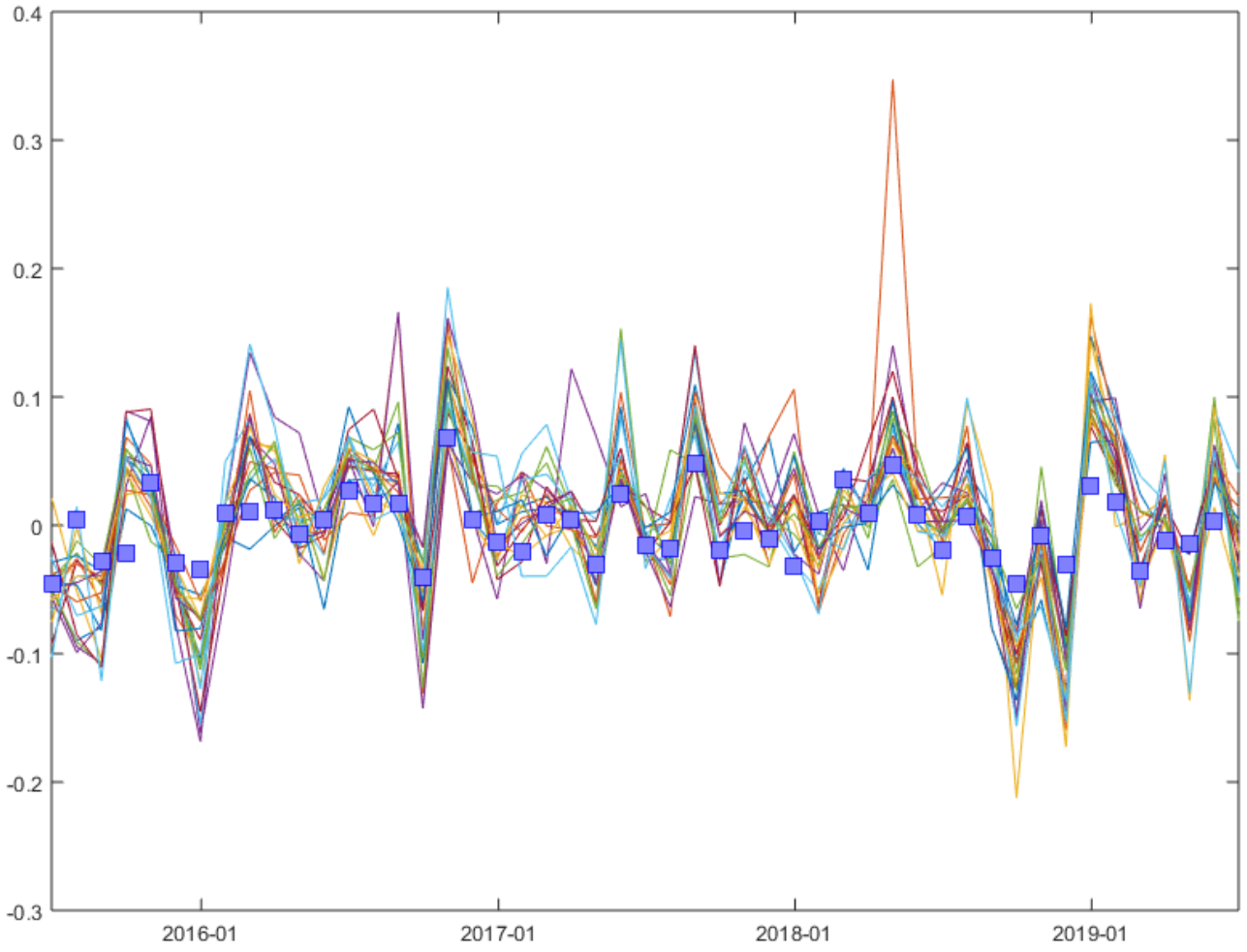

Figure: Fama-French Portfolio Returns

Research Interest

Model specification tests of linear asset pricing models in the presence of weak factors.

Background information: model set up

Illustration of our findings via an example

Weak (proxy) factors distorts the classical HJ test result

Our solutions

I(limited number of asset returns). Robust confidence set

II(extended asset returns). Many-asset approach

Empirical application

Conclusion

Research Interest (Motivation)

- Linear asset pricing models.

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Linear asset pricing model (weak factors)

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a

We want to test the model specification:

Linear asset pricing model (moment conditions)

$$\mathbb{E}\begin{bmatrix} r_{t1} \\ r_{t2} \\ \vdots \\ \vdots \\ r_{tN} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ \vdots \\ 1 \end{bmatrix} \lambda_0 + \begin{bmatrix} \beta_{11} \cdots \beta_{1K} \\ \beta_{21} \cdots \beta_{2K} \\ \vdots \\ \vdots \\ \beta_{N1} \cdots \beta_{NK} \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_K \end{bmatrix}$$

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$$\mathbb{E}\begin{bmatrix} r_{t1} \\ r_{t2} \\ \vdots \\ \vdots \\ r_{tN} \end{bmatrix} f_t' = \begin{bmatrix} \beta_{11} \cdots \beta_{1K} \\ \beta_{21} \cdots \beta_{2K} \\ \vdots \\ \vdots \\ \beta_{N1} \cdots \beta_{NK} \end{bmatrix} V_f$$

x

Linear asset pricing model

$$\mathbb{E}\begin{bmatrix} r_{t1} \\ r_{t2} \\ \vdots \\ \vdots \\ r_{tN} \end{bmatrix} \times \lambda_0^{-1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ \vdots \\ 1 \end{bmatrix} + \begin{bmatrix} \beta_{11} \cdots \beta_{1K} \\ \beta_{21} \cdots \beta_{2K} \\ \vdots \\ \vdots \\ \beta_{N1} \cdots \beta_{NK} \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_K \end{bmatrix} \times \lambda_0^{-1}$$

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$$\mathbb{E}\begin{bmatrix} r_{t1} \\ r_{t2} \\ \vdots \\ \vdots \\ r_{tN} \end{bmatrix} \underbrace{\left(1, f_t'\right)\theta}_{\textit{linear stochastic discount factor } m_t(\theta)} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ \vdots \\ 1 \end{bmatrix}$$

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which can be rewritten as

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$$\mathbb{E}q_t\theta = \iota_N$$

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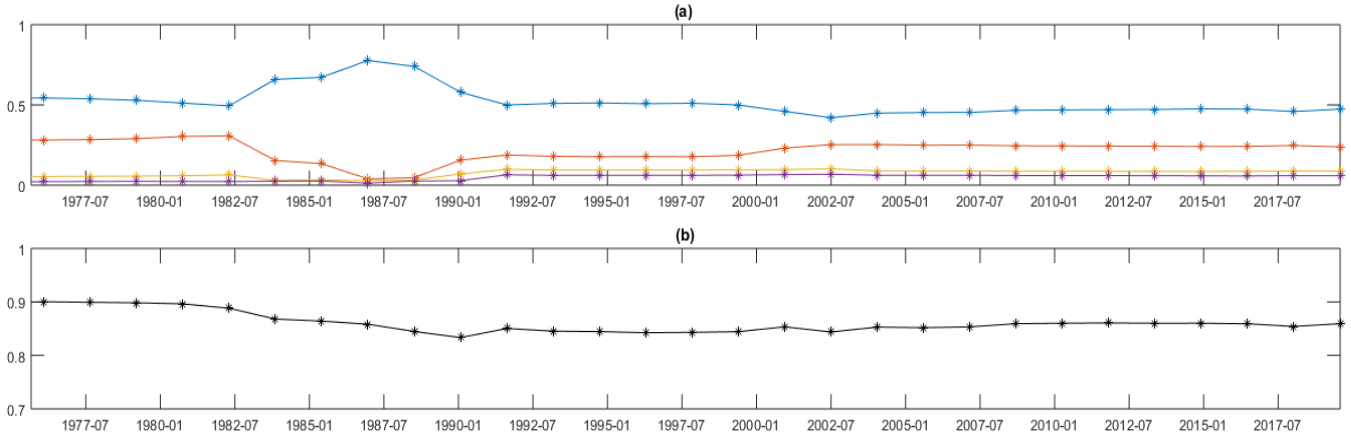

Figure: Fama-French Portfolio Returns

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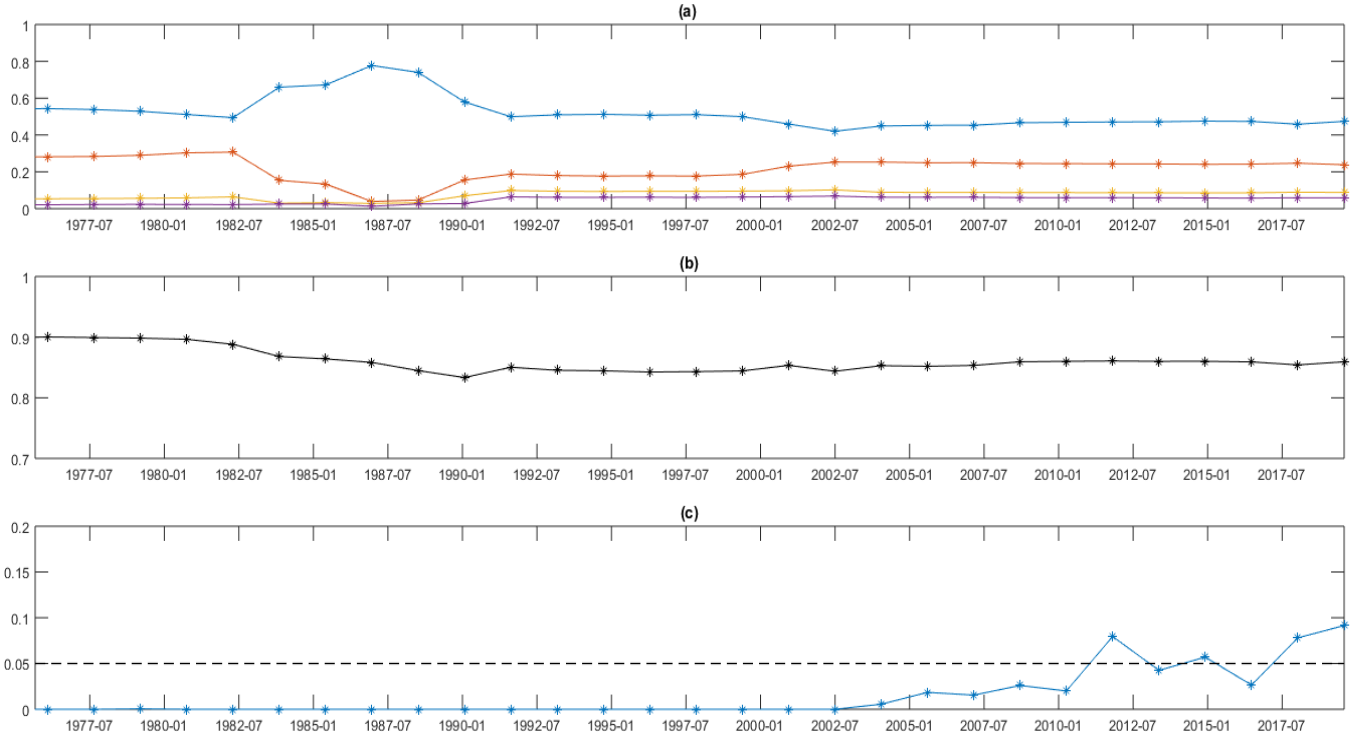

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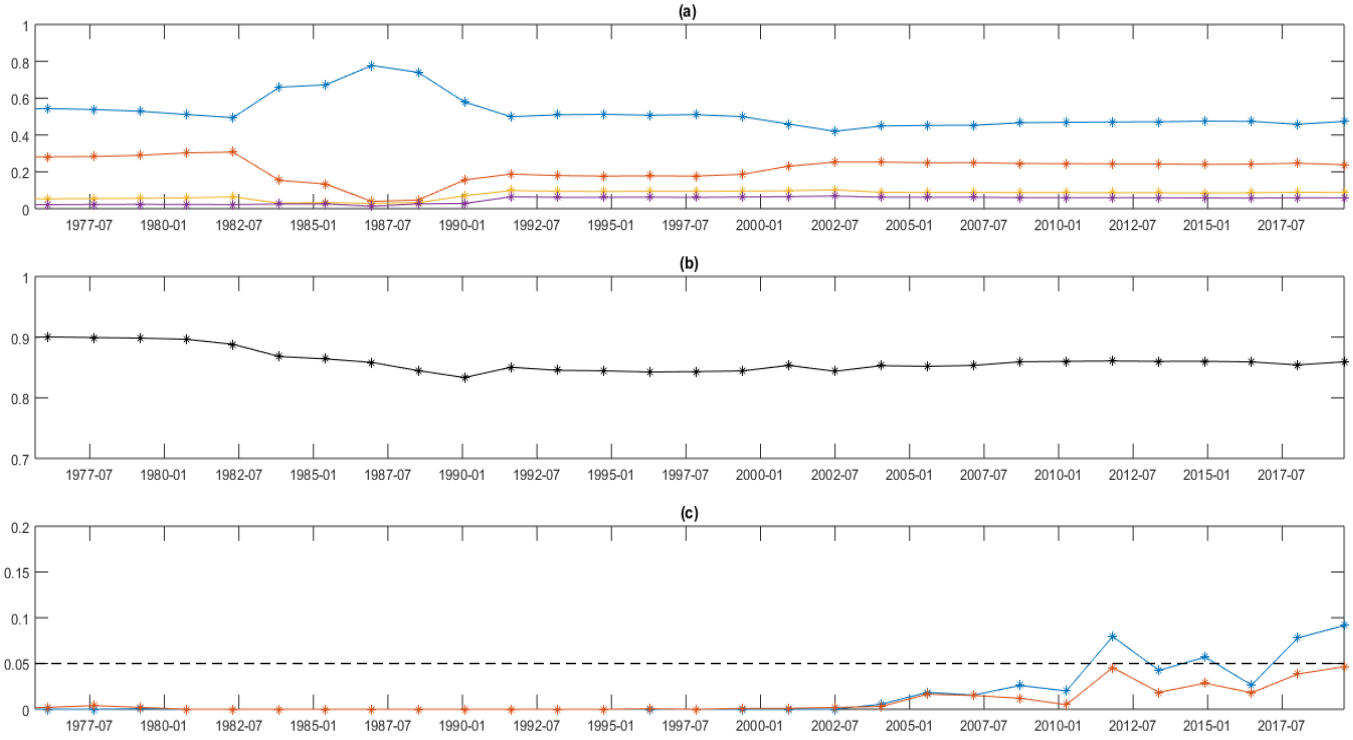

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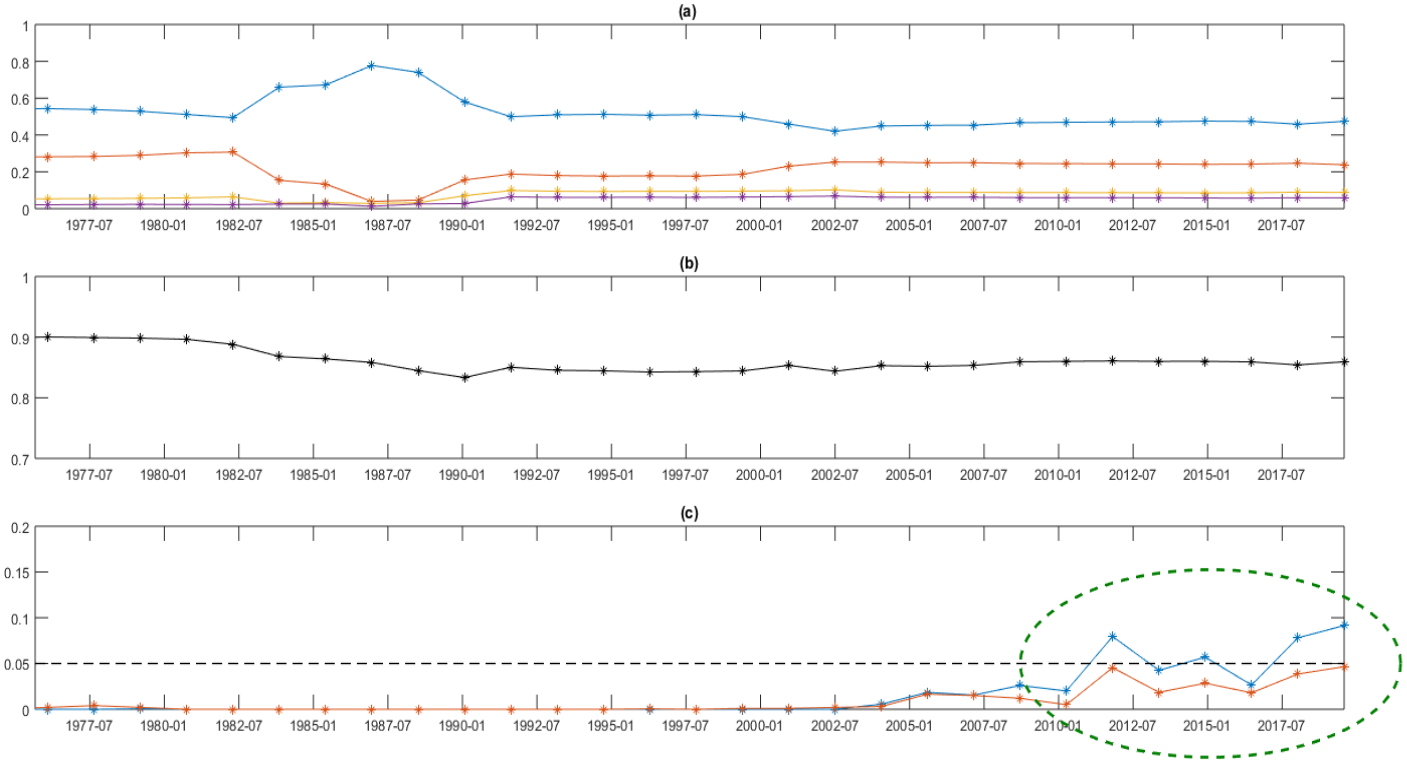

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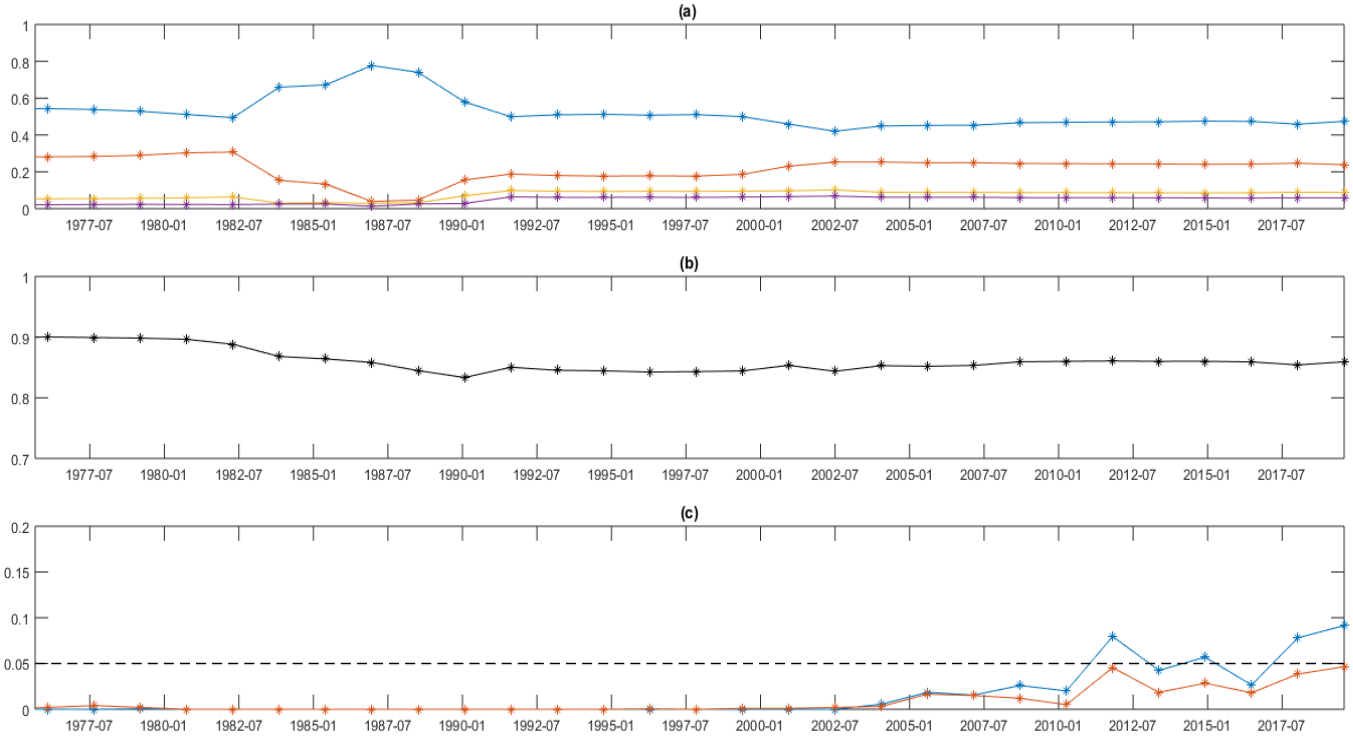

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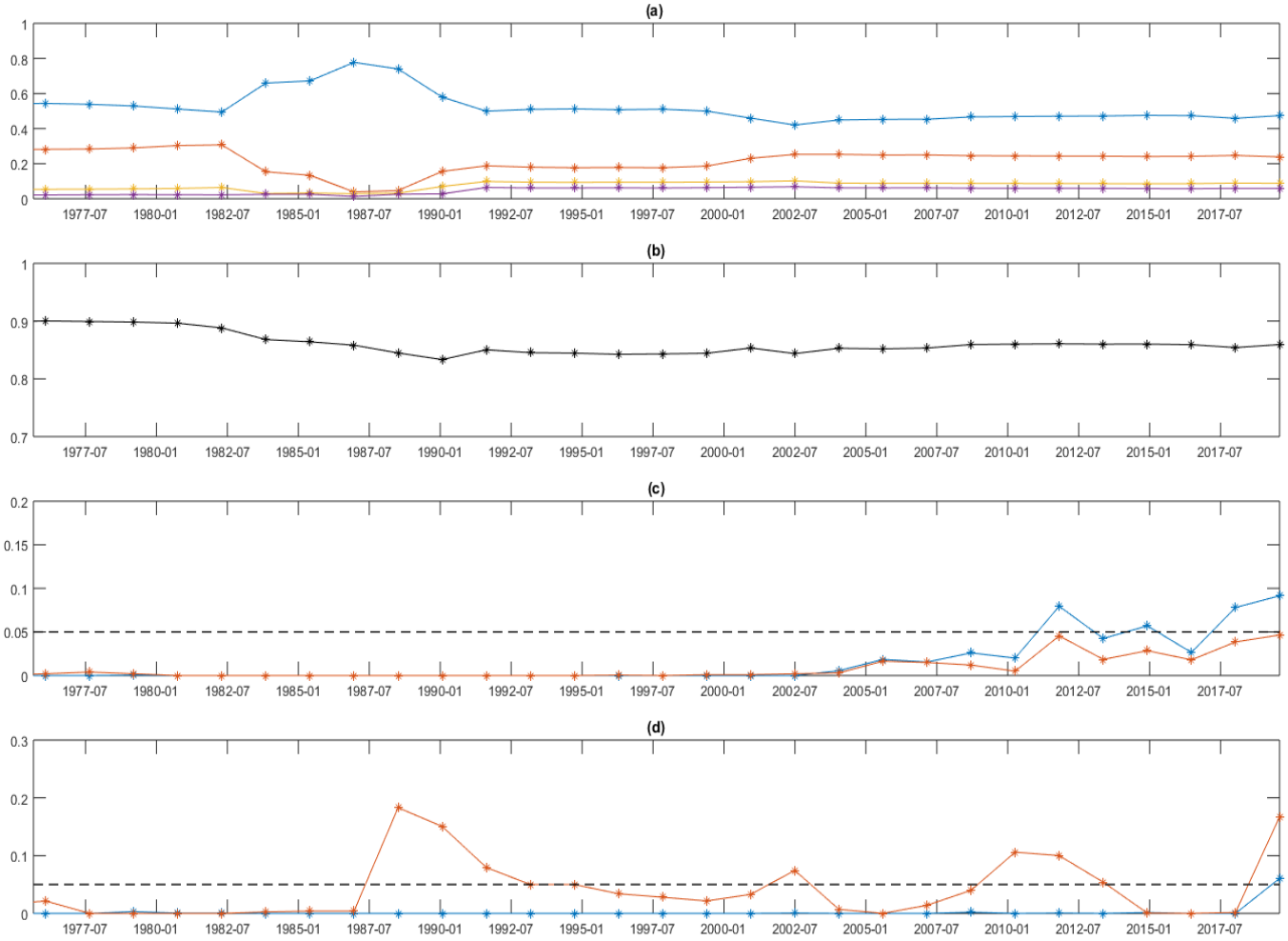

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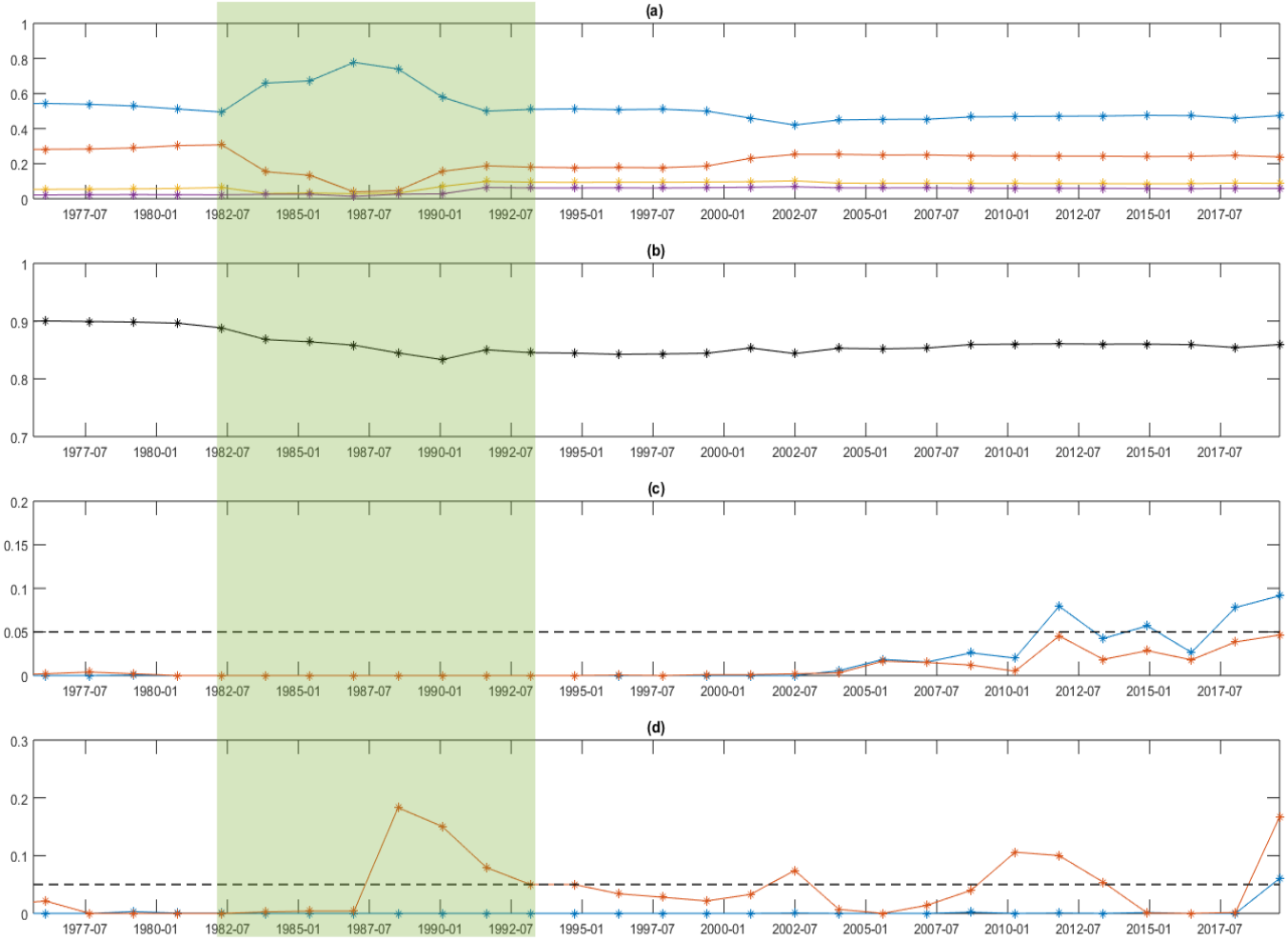

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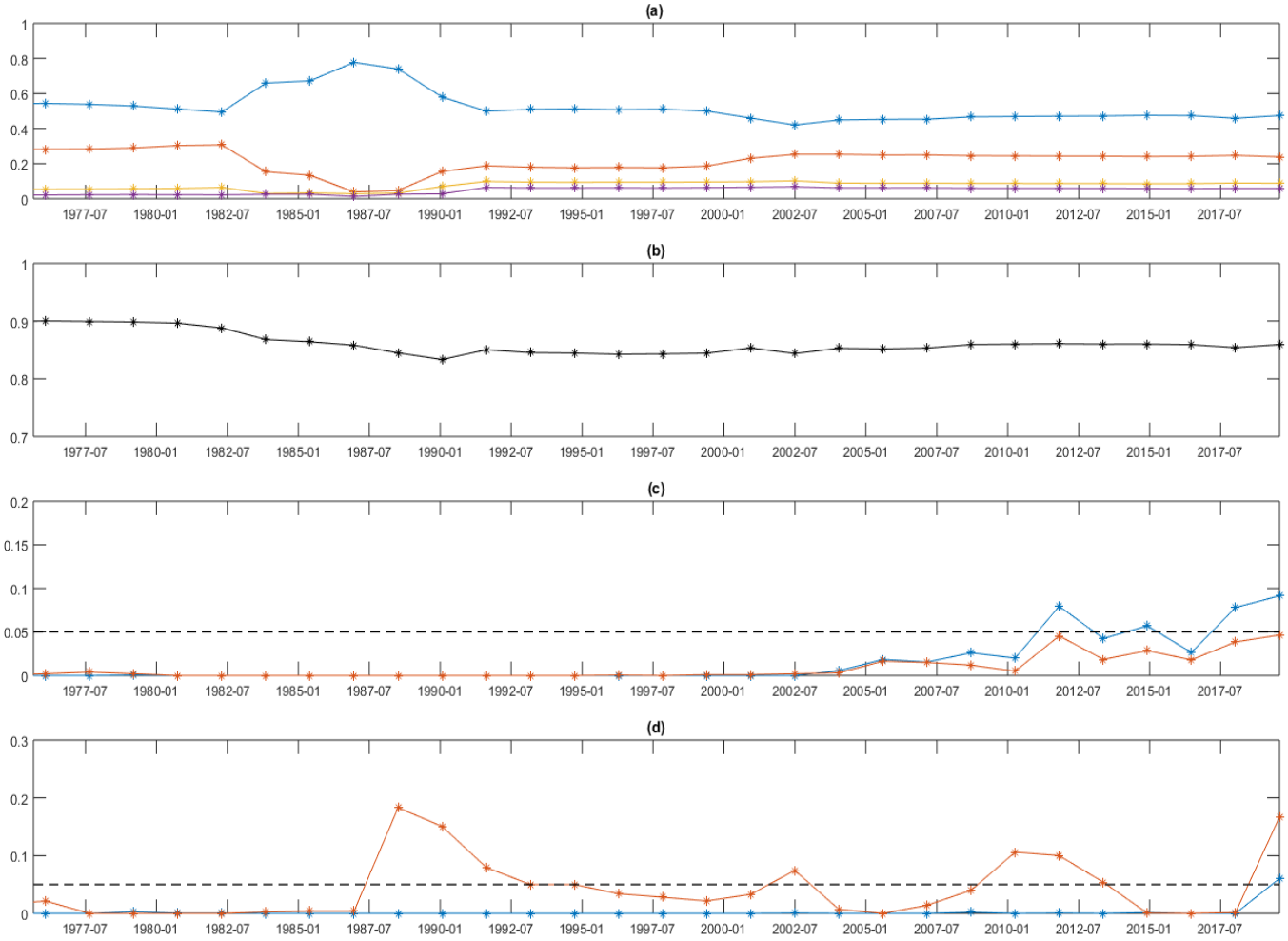

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  - Next, we show that it may be due to the presence of weak (proxy) factors
- We provide alternative specification testing procedures:
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- Besides, an alternative risk premia estimator.

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Linear asset pricing models have two representations: the beta representation and the stochastic discount factor (SDF) representation.

SDF representation (Cochrane (2001)):

$$\begin{aligned} &\mathbb{E}\left(m_t(\theta_0) r_t\right) = \iota_N, \\ &m_t(\theta) = F_t' \theta \end{aligned} \tag{1}$$

with $F_t = (1, f_t')', \theta = (1/\lambda_0, -V_{ff}^{-1}\lambda_f/\lambda_0)$.

HJ distance

The population squared HJ distance δ^2, measuring the deviation from the moment conditions (1), has an explicit expression

$$\delta^2 = \min_{\theta} e(\theta)' Q_r^{-1} e(\theta) = \iota_N' \left(Q_r^{-1} - Q_r^{-1} q \left(q' Q_r^{-1} q \right)^{-1} q' Q_r^{-1} \right) \iota_N,$$

with $e(\theta) = \iota_N - \mathbb{E}\left(m_t(\theta) r_t\right) = \iota_N - q\theta$.

HJ statistic

The HJ statistic, is

$$\widehat{\delta}_g^2 = \min_{\theta_g} \delta_{g,T}^2(\theta_G) = \iota_N' \left(\widehat{Q}_r^{-1} - \widehat{Q}_r^{-1} q_{G,T} \left(q_{G,T}' \widehat{Q}_r^{-1} q_{G,T} \right)^{-1} q_{G,T}' \widehat{Q}_r^{-1} \right) \iota_N,$$

with $\delta_{g,T}^2(\theta_G) = e_{g,T}(\theta_G)' \widehat{Q}_r^{-1} e_{g,T}(\theta_G)$, $e_{g,T}(\theta_G) = \sum_{t=1}^T e_{g,t}(\theta_G)/T = \iota_N - q_{G,T}\theta_G$, $e_{g,t}(\theta_G) = \iota_N - r_t G_t' \theta_G$, $q_{G,T} = \sum_{t=1}^T r_t G_t'/T$, $\widehat{Q}_r = \sum_{t=1}^T r_t r_t'/T$, $G_t = (1, \bar{g}_t')'$.

The estimator resulting from this quadratic optimization problem is

$$\widehat{\theta}_G = \left(q_{G,T}' \widehat{Q}_r^{-1} q_{G,T} \right)^{-1} q_{G,T}' \widehat{Q}_r^{-1} \iota_N.$$

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Simulated critical values

Jagannathan and Wang (1996) propose a model specification test using the **non-pivotal** HJ statistic $T\widehat{\delta}_g^2$. The critical values are drawn from a distribution of weighted sum of independent $\chi^2(1)$ random variables

$$\sum_{i=1}^{N-K_g-1} p_i x_i$$

with x_i being independently $\chi^2(1)$ distributed random variables and p_i being the positive eigenvalues of the matrix

$$S_T^{\frac{1}{2}} \left(\widehat{Q}_r^{-1} - \widehat{Q}_r^{-1} \widehat{B}_g \left(\widehat{B}_g' \widehat{Q}_r^{-1} \widehat{B}_g \right)^{-1} \widehat{B}_g' \widehat{Q}_r^{-1} \right) S_T^{\frac{1}{2}\prime}$$

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Theorem ($\widehat{\theta}_G$ is no longer $\sqrt{T}-$consistent)

Suppose all proxy factors g_t are weak, then

$$\left(\widehat{V}_g(\widehat{\theta}_g - \theta_g) - bias_e - bias_m \right) = O_p(1)$$

Model selection failures

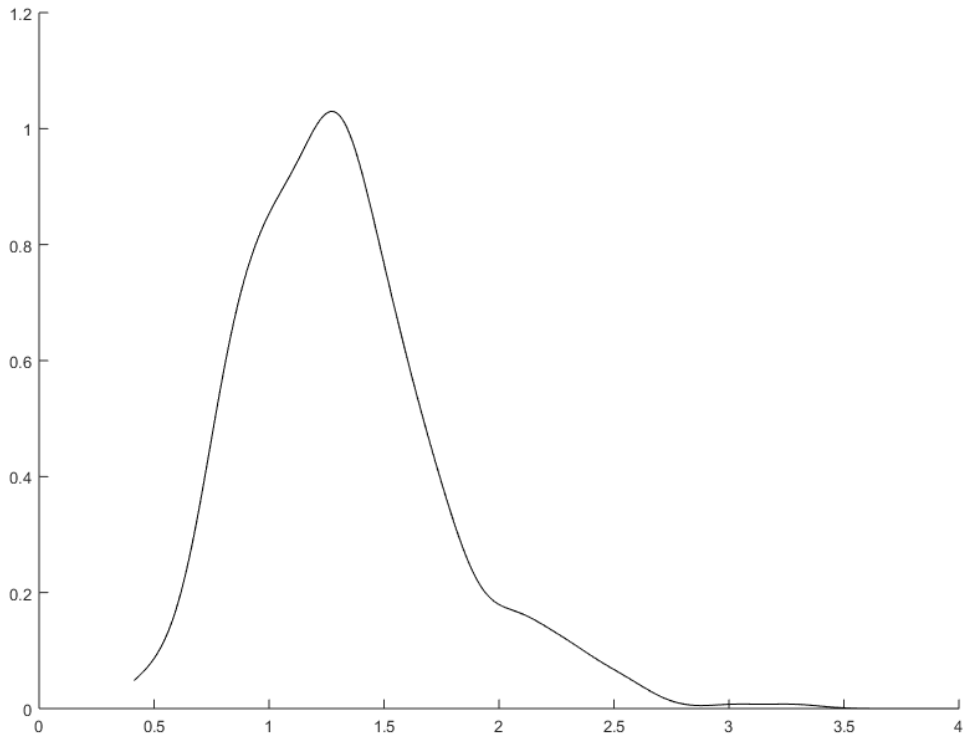

Figure: Density functions of the HJ statistic: (1) black solid: three strong factors; (2) black dashed: two strong factors; (3) black dotted: one strong factors; (4) blue solid: three weak proxy factors; (5) red solid: three strong factors and one useless factor; (6) red dashed: two strong factors and one useless factor; (7) red dotted: one strong factors and one useless factor; (8) red dash-dotted:one strong factors and two useless factors

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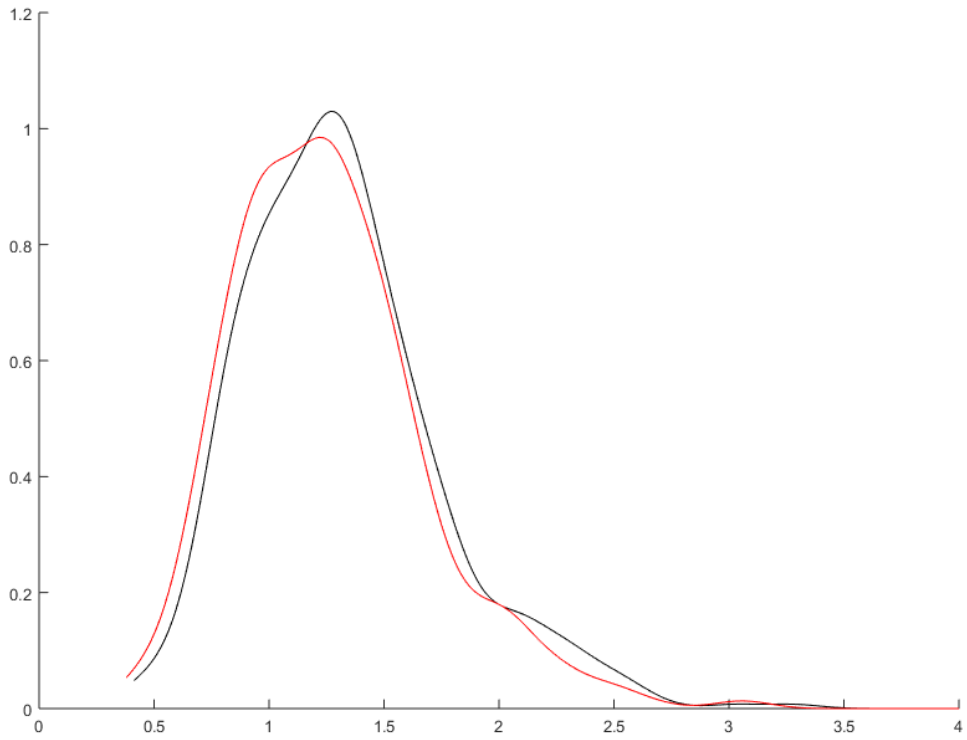

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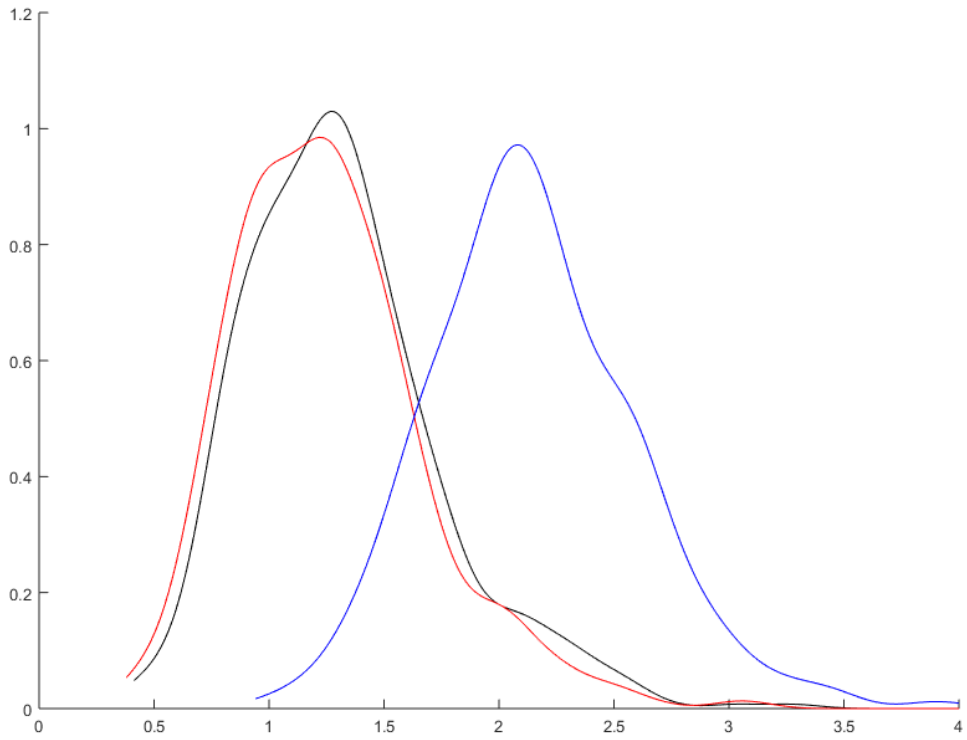

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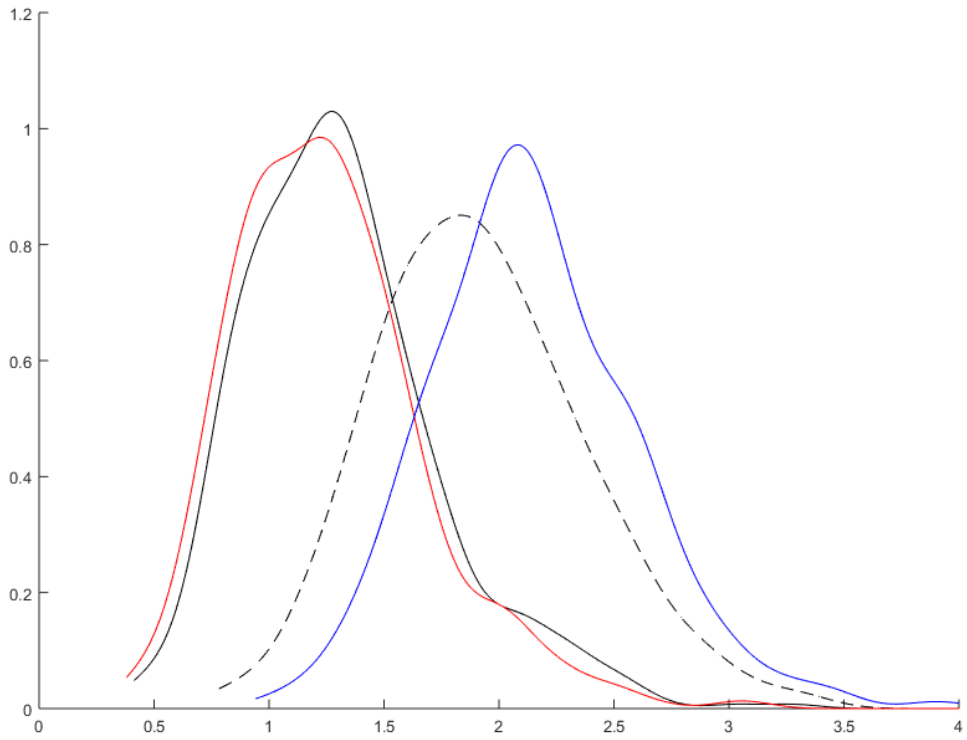

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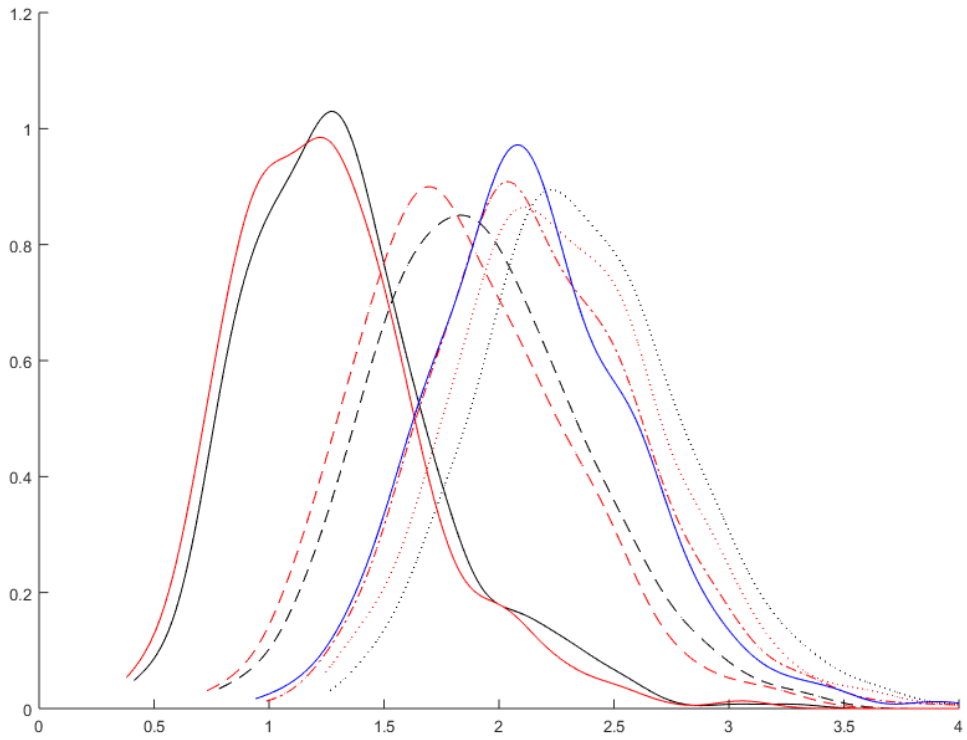

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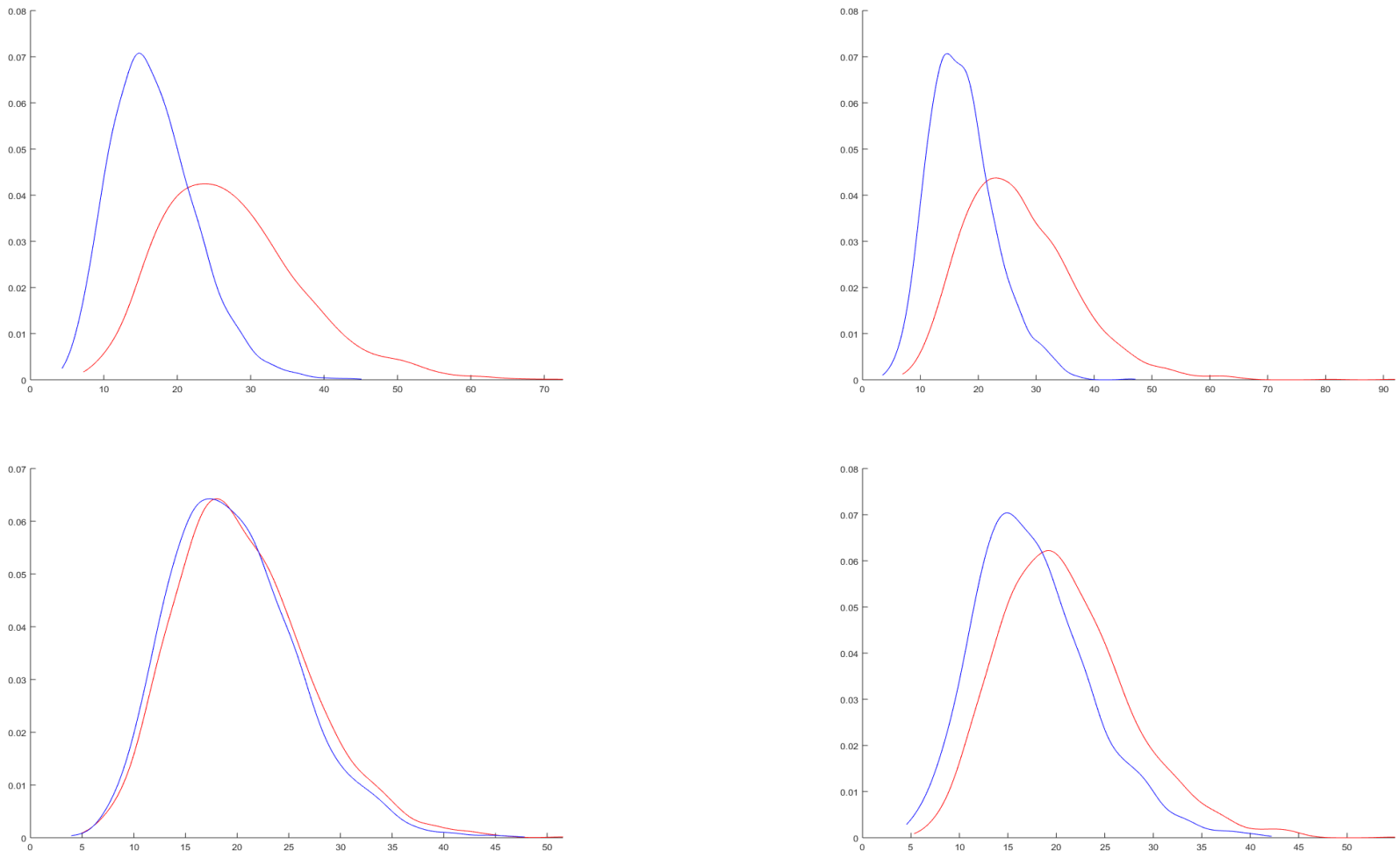
Specification testing failures



Density functions of the HJ statistic (red solid curve), weighted sum of χ^2's approximation (blue solid curves), asymptotic distribution based on Theory 21 (dot-dashed curve). Top left-hand side panel: $T = 100$, three strong factors f_t; top right-hand side panel: $T = 100$, three weak factors g_t; bottom left-hand side panel: $T = 1000$, three strong factors f_t; bottom right-hand side panel: $T = 1000$, three weak factors g_t.

Theorem (asymptotic distribution of the HJ statistic)

Under some mild assumptions, let T increase to infinity with fixed N then the behavior of the HJ statistic is characterized by:

$$T\widehat{\delta}_g^2 \to_d \widetilde{\psi}'_{B_g} M_{Q_r^{-\frac{1}{2}}\left(\eta_{B_g} + (0 \vdots \psi_{\beta_{g,2}})\right)} \widetilde{\psi}_{B_g}$$

with $\widetilde{\psi}_{B_g} = \psi_{B_g} Q_G \theta_G$.

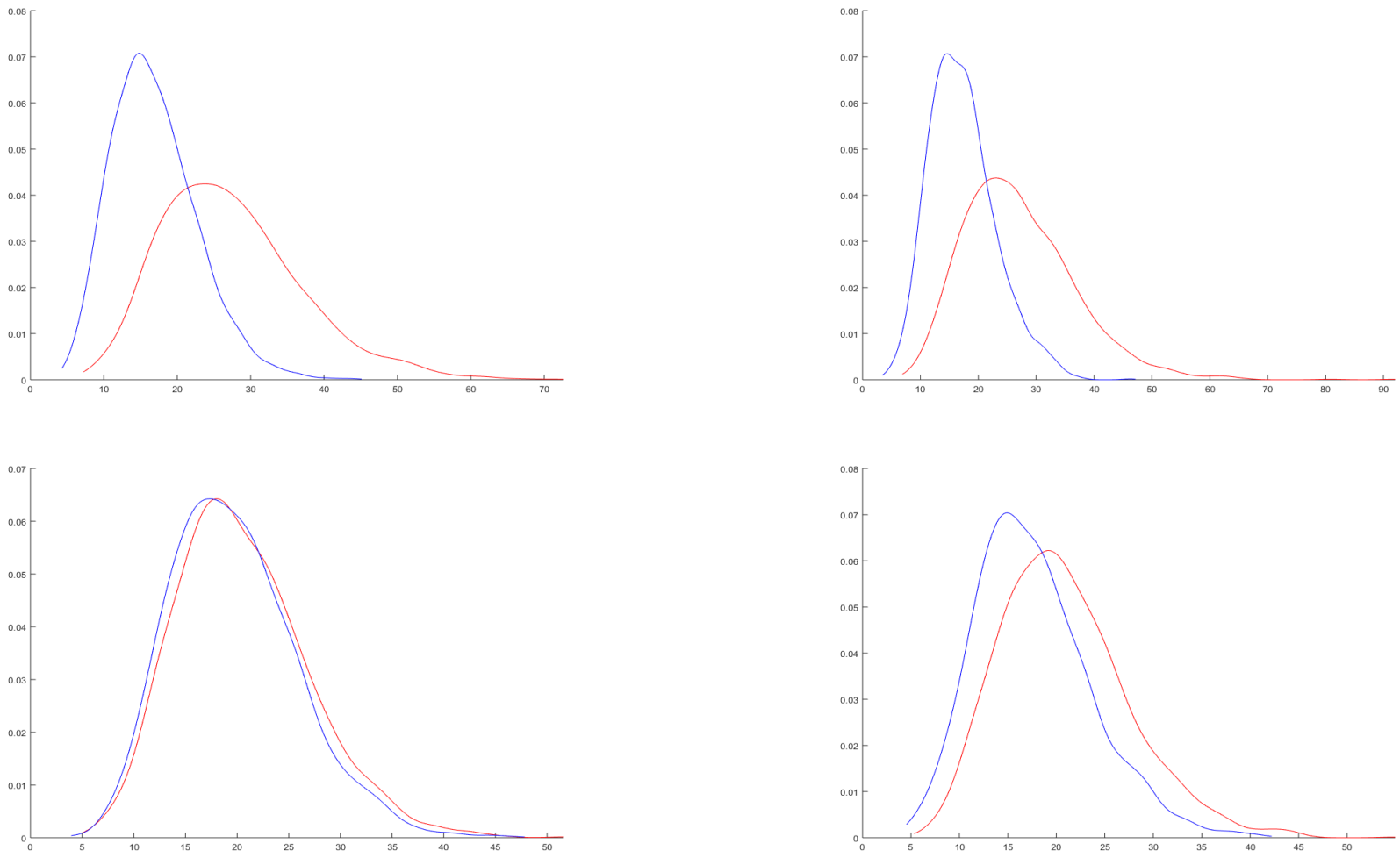
Theorem (size distortion of the HJ specification testing procedure)

Under some mild assumptions, in the presence of weak (proxy) factors

$$\limsup_N \limsup_T \mathbb{P}_0\left(T\widehat{\delta}_g \geq \widehat{c}_{1-\alpha}\right) = 1$$

where $\widehat{c}_{1-\alpha}$ is the conventional critical value.

Specification testing failures



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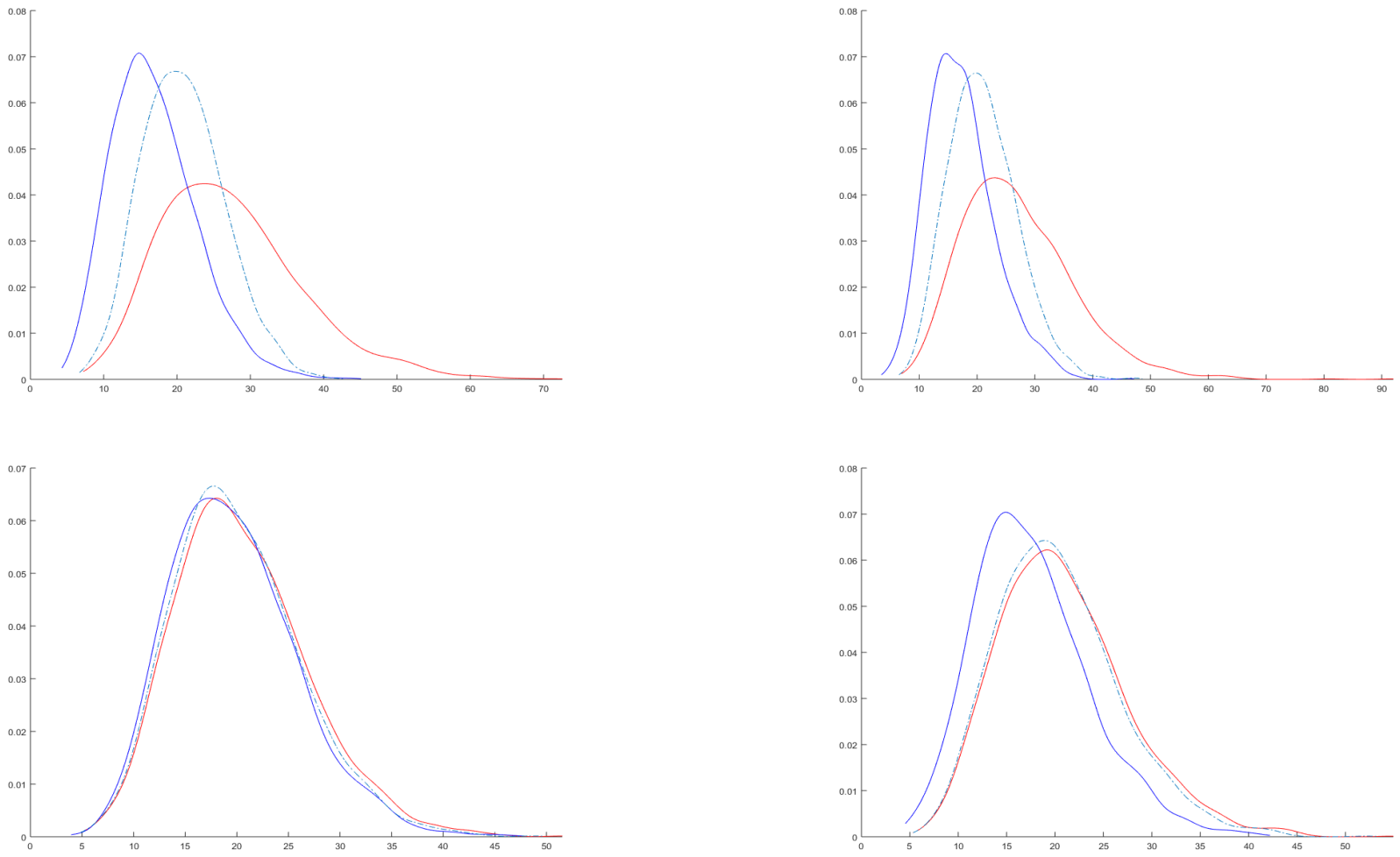

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β	$N = 5$	$N = 10$	$N = 15$	$N = 31$
T=100, $d_g = 1.9$	0.5032	0.8824	0.9711	0.9992
T=10000, $d_g = 1.9$	0.1210	0.1486	0.2298	0.2238
T=100, $d_g = 0.9$	0.7132	0.9330	0.9814	1
T=10000, $d_g = 0.9$	0.5174	0.8906	0.8834	0.9978

Table: Rejection frequency table under the null (at 0.05 significance level) via the $T\widehat{\delta}_g$ with critical values drawn from weighted sum of χ^2's

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HJS statistic and critical values

$$\widehat{\delta}^* = \inf_{\theta \in CS_{r,\alpha_1}} \delta(\theta) \tag{2}$$

with CS_r is the robust confidence set constructed by inverting the AR test statistic

$$CS_{r,\alpha_1} = \left\{\theta \in \Theta : AR(\theta) \leq c_{1-\alpha_1}\right\}; \quad AR(\theta) = Te_T(\theta)'\widehat{\Sigma}^{-1}_{e_T(\theta)} e_T(\theta)$$

and the critical value of $T\widehat{\delta}^*$ is constructed as

$$c^*_{1-\alpha} = \sup_{\theta \in CS_{r,\alpha_1}} c^*_{1-\alpha_2}(\theta) \tag{3}$$

with $c^*_{1-\alpha}(\theta)$ the critical values drawn from weighted sum of N $\chi^2(1)$ random variables with weights being the non-zero eigenvalues of $(\widehat{\Sigma}_e(\theta))^{\frac{1}{2}}\,\widehat{Q}_r^{-1}(\widehat{\Sigma}_e(\theta))^{\frac{1}{2}\,\prime}$ and α_1, α_2 is chosen such that $(1-\alpha_1)(1-\alpha_2) = 1-\alpha$.

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Lemma (confidence set coverage probability)

Suppose some regular assumptions hold, then even in the presence of weak identification

$$\liminf_T \mathbb{P}\left(\theta_G \in CS_{r,\alpha}\right) \geq 1-\alpha$$

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Theorem (the HJS specification test is size correct)

Suppose some regular assumptions hold, then even in the presence of weak identification

$$\limsup_T \mathbb{P}\left(T\widehat{\delta}^*_g \geq c^*_{1-\alpha}\right) \leq \alpha$$

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Sketch of proof: $\mathbb{P}\left(T\widehat{\delta}^*_g \geq c^*_{1-\alpha}\right) \leq \mathbb{P}\left(T\widehat{\delta}_g(\theta_G) \geq c^*_{1-\alpha}\right) \leq \mathbb{P}\left(T\widehat{\delta}_g(\theta_G) \geq c_{1-\alpha}(\theta_G)\right)$

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Example (sup does not work)

Suppose some regular assumptions hold, $K = K_{g,2} \geq 2$, then

$$\limsup_T \mathbb{P}\left(\left(\sup_{\theta \in CS_{r,\alpha_1}} \delta_{g,T}(\theta)\right) \geq c^*_{1-\alpha}\right) = 1$$

Power comparison: HJS vs J

$\mathcal{J}$ specification test

$$\mathcal{J} = \inf_{\theta} AR(\theta)$$

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Theorem: case (i) (*Gospodinov et al. 2017* unidentified model under misspecification)

Suppose some regular conditions hold. Let $H = (\iota_N, q_G)$. Suppose H has a column rank $K + 1 - k$ for an integer $k \geq 1$, then we have

$$\mathcal{J} \to_d w_{k,i}$$

where $w_{k,i}$ is the smallest eigenvalue of $\mathcal{W}_k \sim \mathcal{W}_k\left(N - K - 1 + k, I_k\right)$ and $\mathcal{W}_k\left(N - K - 1 + k, I_k\right)$ denotes the Wishart distribution with $N - K - 1 + k$ degrees of freedom and a scaling matrix I_k. Furthermore,

$$\limsup_{T} \mathbb{P}\left(\mathcal{J} \geq c_{\chi^2_{N-K},1-\alpha}\right) \leq \alpha,$$
$$\liminf_{T} \mathbb{P}\left(T\widehat{\delta}^*_g \geq c^*_{1-\alpha}\right) = 1,$$

with $c_{\chi^2_{N-K},1-\alpha}$ the $1 - \alpha$ quantile of χ^2_{N-K}.

Power comparison: HJS vs J

$\mathcal{J}$ specification test

$$\mathcal{J} = \inf_{\theta} AR(\theta)$$

Theorem: case (ii) (*weakly identified model under misspecification*)

Suppose some regular conditions hold. Let $H = \left(\iota_N, q_G Q_G^{-1}\right)$. Suppose Suppose $(HQ_x)' * (HQ_x)$ with $Q_x = diag(I, \sqrt{T} I_{K_{g,2}})$ converges to a positive definite matrix, then we have

$$\mathcal{J} \to_d w_{k,ii}$$

where $w_{k,ii}$ is the smallest eigenvalue of $\mathcal{W}_k \sim \mathcal{W}_k\left(\mu, N-K-1+k, I_k\right)$ and $\mathcal{W}_k\left(\mu, N-K-1+k, I_k\right)$ denotes the non-central Wishart distribution with $N-K-1+k$ degrees of freedom and a scaling matrix I_k, a location parameter μ (μ is specified in the proof). Furthermore,

$$\liminf_T \mathbb{P}\left(\mathcal{J} \geq c_{\chi^2_{N-K},1-\alpha}\right) < 1,$$
$$\liminf_T \mathbb{P}\left(T\widehat{\delta}_g^* \geq c^*_{1-\alpha}\right) = 1,$$

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$\widehat{\theta}_G$ revisit

$$\iota_N = q_{G,T}\theta_G - \epsilon_{q_G}\theta_G,$$

with $\epsilon_{q_G} = (q_{G,T} - \mathbb{E}q_{G,T})$ being correlated with the $q_{G,T}$.

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Four-pass estimator

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Step (1): Estimate $\widehat{c}, \widehat{\beta}_g$ in by regressing r_t on a constant and proxy factors.
Step (2): Determine the omitted factor structure
by the following two steps: (2.1) determine the number of factors, K_{vz}, in $\widehat{u}_{g,t} = r_t - \widehat{c} - \widehat{\beta}_g \bar{g}_t$ by

$$\widehat{K}_{vz} = \arg \min_{0 \le j \le K_{vz,\max}} \left(N^{-1} T^{-1} \lambda_j \left(\widehat{u}_g \widehat{u}_g' \right) + j\phi(N, T) \right) - 1$$

where $\widehat{u}_g$ is $T \times N$ matrix stacked with the OLS residuals $\widehat{u}_{g,t}$, $K_{vz,\max}$ is an arbitrary upper bound for K_{vz} and $\phi(N, T)$ is a penalty function with the properties
$\phi(N, T) \to 0, \phi(N, T)/(N^{-\frac{1}{2}} + T^{-\frac{1}{2}}) \to \infty$; (2.2): estimate the $T \times N$ common component matrix $cc = xb'$ stacked with the common components cc_t:

$$\widetilde{cc} = \widehat{x}\widehat{b}, \quad \left(\widehat{x}, \widehat{b}\right) = \arg \min_{b_i, x_t \ s.t. \ \sum_{t=1}^{T} x_t x_t' / T = I_{\widehat{K}_{vz}}} \sum_{i,t} \left(\widehat{u}_{g,it} - b_i' x_t \right)^2$$

The estimator $\widehat{x}$ is equal to the $\sqrt{T}$ times eigenvector associated with the $\widehat{K}_{vz}$ largest eigenvalues of the matrix $\widehat{u}_g \widehat{u}_g'$, and $\widehat{b} = \widehat{x}' \widehat{u}_g / T$ corresponds to the OLS estimator regressing $\widehat{u}_g$ over $\widehat{x}$.

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Step (3): Remove the missing factor structure from regressors

$$\widetilde{q}_{G,T}^{(i)} = q_{G,T}^{(i)} - \widetilde{cc}_{G,T}^{(i)\prime}, i = 1, 2$$

where

$$q_{G,T}^{(1)} = \frac{1}{\lfloor \frac{T}{2} \rfloor} \sum_{t=1}^{\lfloor \frac{T}{2} \rfloor} r_t G_t'; \; q_{G,T}^{(1)} = \frac{1}{T - \lfloor \frac{T}{2} \rfloor} \sum_{t=\lfloor \frac{T}{2} \rfloor + 1}^{T} r_t G_t'$$

$$\widetilde{cc}_{G,T}^{(2)} = \frac{1}{\lfloor \frac{T}{2} \rfloor} \sum_{t=1}^{\lfloor \frac{T}{2} \rfloor} \widetilde{cc}_t G_t'; \; \widetilde{cc}_{G,T}^{(2)} = \frac{1}{T - \lfloor \frac{T}{2} \rfloor} \sum_{t=\lfloor \frac{T}{2} \rfloor + 1}^{T} \widetilde{cc}_t G_t'.$$

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with $\widetilde{\theta}_G^{(1)} = \left(\widetilde{q}_{G,T}^{(1)\prime} P_{\widetilde{q}_{G,T}^{(2)}} \widetilde{q}_{G,T}^{(1)} \right)^{-1} \widetilde{q}_{G,T}^{(1)\prime} P_{\widetilde{q}_{G,T}^{(2)}} \iota_N$ and $\widetilde{\theta}_G^{(2)} = \left(\widetilde{q}_{G,T}^{(2)\prime} P_{\widetilde{q}_{G,T}^{(1)}} \widetilde{q}_{G,T}^{(2)} \right)^{-1} \widetilde{q}_{G,T}^{(2)\prime} P_{\widetilde{q}_{G,T}^{(1)}} \iota_N$.

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Theorem (consistency)

Under regular assumptions and $N/T \to c \geq 0$

$$\sqrt{NT} Q_x^{-1} V_g \left(\widetilde{\theta}_g - \theta_g\right) \to N(0, \Sigma_{\theta_g})$$

with $Q_x = \text{diablk}\left(1, I_{K_{g,s}}, \sqrt{T} I_{K_{g,w}}\right)$. When $c = 0$ we have one estimator for Σ_{θ_g}.

Alternative specification testing procedure: HJN statistic

The HJN statistic

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Asymptotic characterization

$$T\widetilde{\delta}_g^2 = T\left(\iota_n - \widetilde{q}_G\widetilde{\theta}_G\right)' \widehat{Q}_R^{-1} \left(\iota_n - \widetilde{q}_G\widetilde{\theta}_G\right)$$

$$\sqrt{T}\left(\iota_n - \widetilde{q}_G\widetilde{\theta}_G\right) = \underbrace{\sqrt{T}\left(\iota_n - \widetilde{q}_G\theta_G\right)}_{O_p(1)} - \underbrace{\sqrt{T}\left(\widetilde{q}_G\left(\widetilde{\theta}_G - \theta_G\right)\right)}_{O_p(\frac{1}{N^{1/2}})}$$

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Theorem (HJN specification test)

Under regular assumptions,

$$T\widetilde{\delta}_g^2 \sim_d \sum_{i=1}^{n} p_i x_i$$

with x_i being independently $\chi^2(1)$ distributed random variables and p_i being the positive eigenvalues of the matrix $S^{\frac{1}{2}}(\widehat{Q}_R)^{-1}S^{\frac{1}{2}\prime}$ with
$S = \frac{1}{T}\sum_{t=1}^{T} e_{g,t,R}(\widetilde{\theta}_G)e_{g,t,R}(\widetilde{\theta}_G)'$, $e_{g,t,R}(\theta_g) = \iota_N - R_t G_t'\theta_g$.

Alternative specification testing procedure: HJN statistic

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Research Interest

Model specification tests of linear asset pricing models in the presence of weak factors.

Background information: model set up

Illustration of our findings via an example

Weak (proxy) factors distorts the classical HJ test result

Our solutions

I(limited number of asset returns). Robust confidence set

II(extended asset returns). Many-asset approach

Empirical application

Conclusion

Empirical application

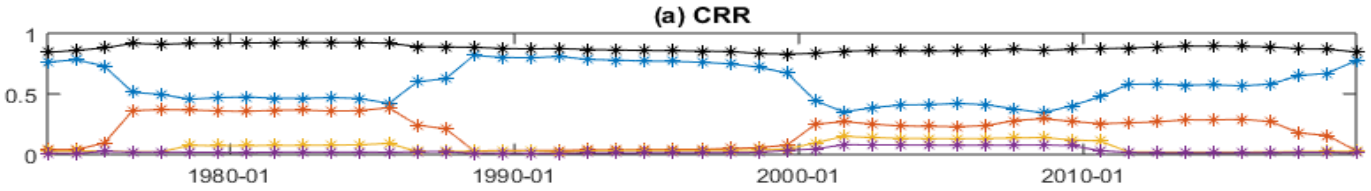
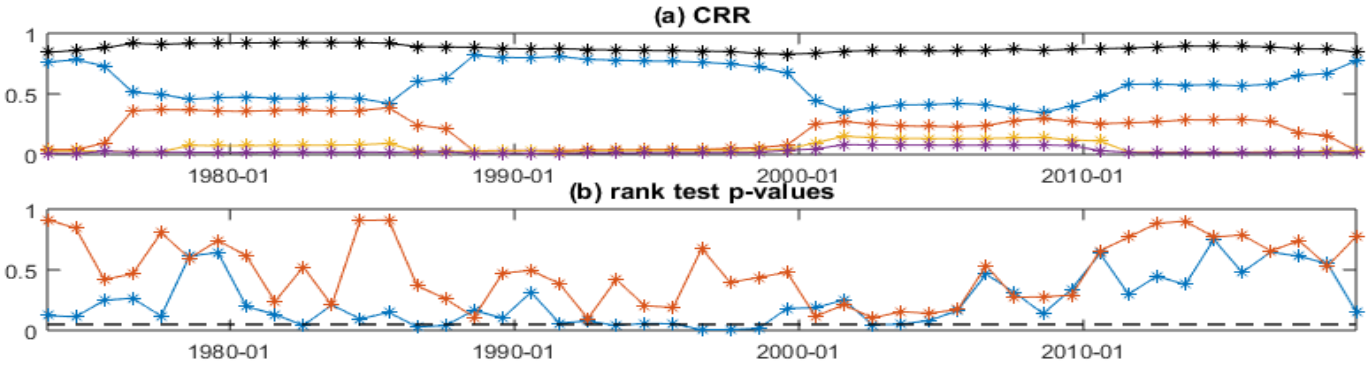


Figure: Fama-French Portfolio Returns

Empirical application



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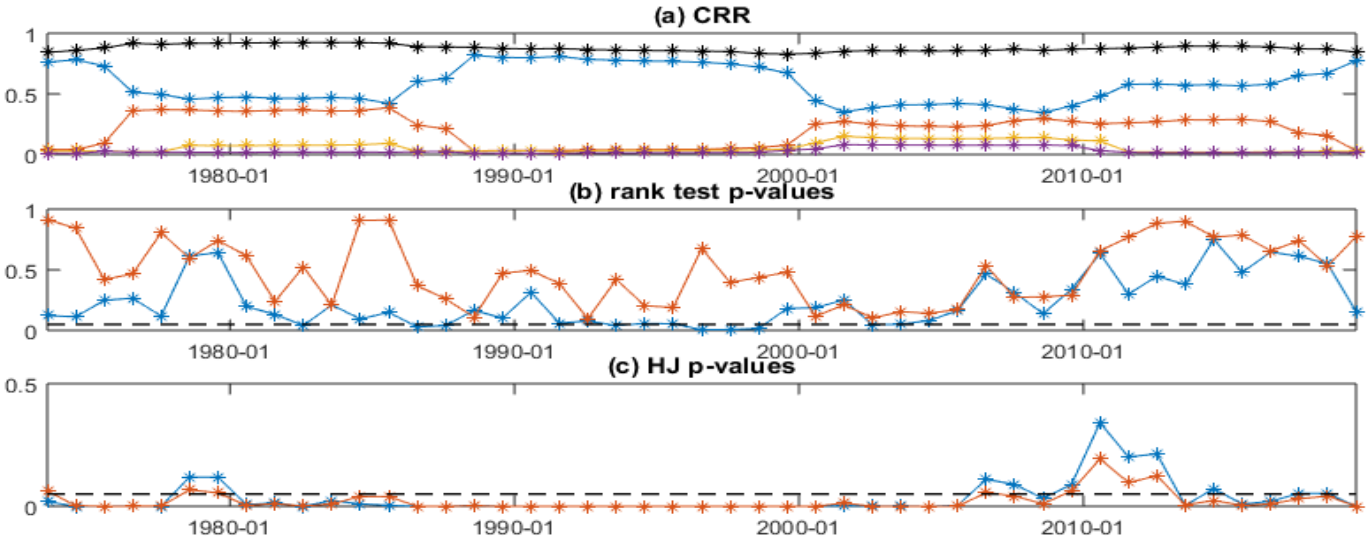

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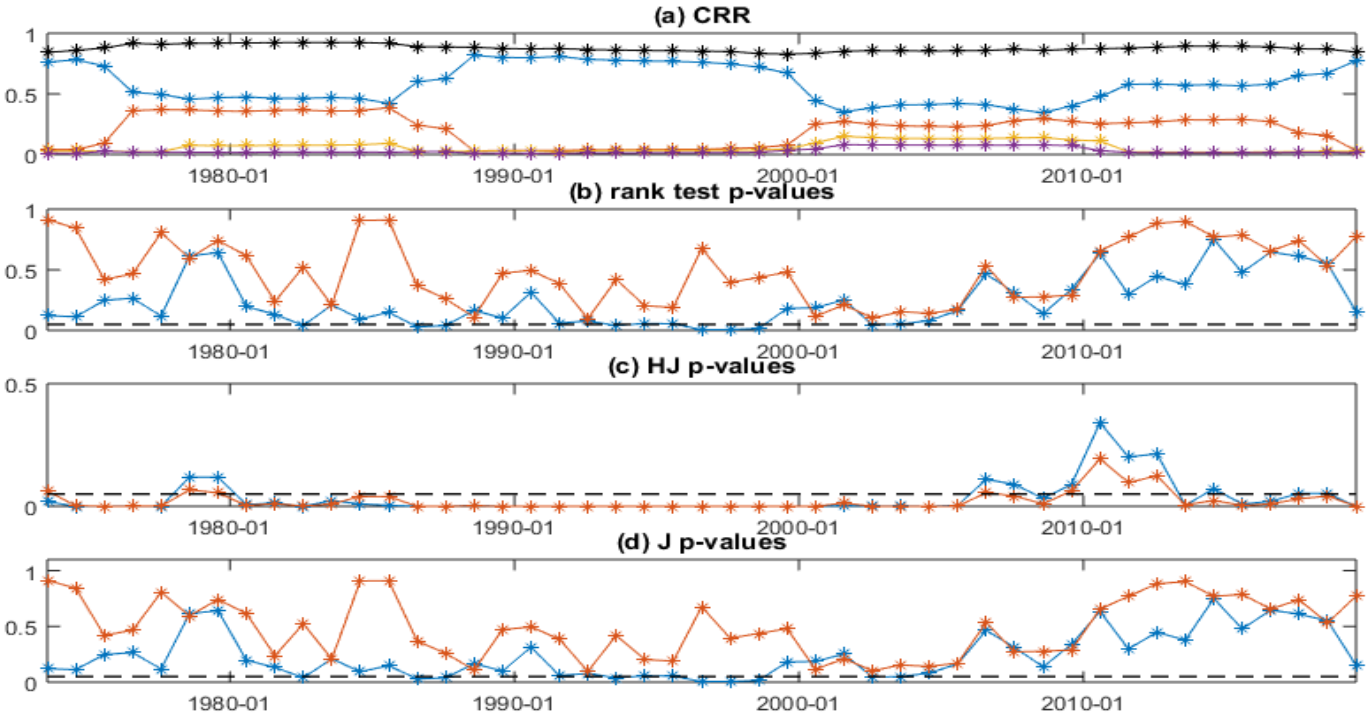

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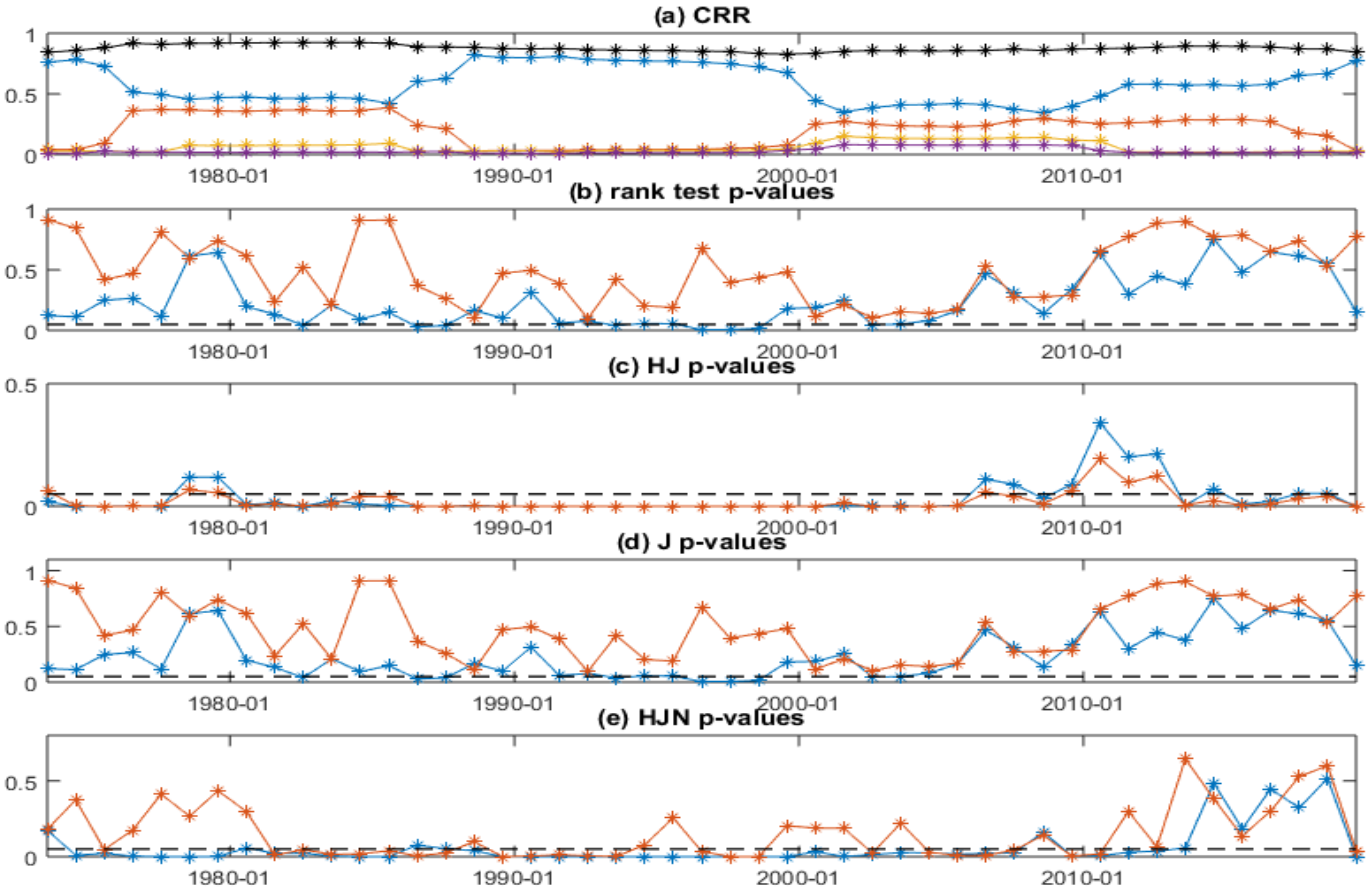

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Empirical application

	Market	SMB	HML	MOM	HJ(p-val)	HJN(p-val)	HJS(rejected)
(1)					0.000	0.000	Yes
$\widehat{\theta}_G$	0.0182	-0.0313	-0.0946	–			
$\widetilde{\theta}_G$	0.2467	-0.1885	0.1258	–			
$\widehat{\lambda}_g$	0.0602	0.5279	0.3742	–			
$\widetilde{\lambda}_g$	-3.2179	0.6038	0.0347	–			
(2)					**0.001**	**0.514**	No
$\widehat{\theta}_G$	0.0248	-0.0239	-0.0592	0.1952			
$\widetilde{\theta}_G$	0.1812	0.0404	0.2959	0.8543			
$\widehat{\lambda}_g$	0.3387	0.4694	0.3730	3.1841			
$\widetilde{\lambda}_g$	-3.3109	0.3053	0.0226	-10.7839			

Table: Estimates with Fama-French factors and the momentum factor and tests of specification using monthly returns on 100 portfolios sorted by size and book-to-market from 1963 to 1998

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- We document failures of conventional approaches when using the HJ distance statistic
- We provide novel approaches for carrying out identification robust specification test
- We provide a novel risk premia estimator.